Abhishek Singh

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Areas of Interests

Natural Language Processing, Code-mixed languages, Natural Language Understanding

Academic History

Year	Qualification	Institute	Performance	Subjects
2015-2019	B.Tech, Electrical Engineering	Indian Institute of Technology (B.H.U), Varanasi	9.09/10	Project-Localization of Mobile Robots
2015	XII (C.B.S.E)	D.A.V Public School, Kota	92.2/100	Mathematics-97/100
2013	X (I.C.S.E)	St. Peter's School, Panchgani	91.5/100	Computer Science- 98/100

Skills and Certifications

- **Programming/ Scripting Languages**- C, C++, Python, Java, CSS, PHP, HTML,SQL
- **Software/Libraries**- PyTorch, Tensorflow, Keras, Librosa, scikit-learn, CUDA, Git
- **Machine Learning**-Natural Language Processing, Sentiment analysis, Text Summarization, text classification, Reinforcement learning
- **MOOCs**- Deep Learning Specialization, Sequence Models, CNNs, Hyper-Parameter Tuning, Optimization, Structuring Machine Learning Projects, Neural Network, and Deep Learning, Machine Learning-Coursera

Job, Research, Teaching and Internship Experience

Job Experience

Samsung R&D Bangalore (Speech recognition and Natural Language Processing Engineer) **June 2019-Present**

- Working on seamless conversation technology for Virtual Assistant. Detecting device and non-device directed speech.
- Working on language model to train Voice assistant on code-mixed Hinglish (Hindi+English) language.
- Took part in SemEval2020 tasks, OffensEval for offensiveness detection on multilingual data and Sentimix for sentiment analysis on code-mixed Hinglish and Spanglish (Spanish+English) languages.

Research Internship

Design and Innovation Hub at IIT(BHU) Varanasi **Prof.R.K.Misra** **May-July 2017**

- Developed Solar Energy Forecasting model using Tensorflow, applied LSTM and Ridge Regression, with an accuracy of 16% MAPE.
- Applied Random Forest, LightGBM and Ridge Regression on the dataset from Kaggle Solar Forecast Challenge to compare accuracy.

Design and Innovation Hub at IIT(BHU) Varanasi **Prof.Devender Singh** **May-July 2016**

- Designed a Smart Metering System, using GSM and Arduino UNO, and created a website for Electricity Users for Usage analysis.
- Features of this model are Online Monitoring through webpage, Electricity Cut-off and Electric Billing through an SMS using GSM.

Teaching Assistant

1. Basic Electrical Engineering- Teaching and Paper Correction (Under Graduate Sophomore Level Course) **July- December 2018**
2. Computer Programming and Linux Concepts-Teaching and Paper Correction (UG freshman Level Course) **January-May 2018**

Software Engineering Internship at Indus Valley Partners, Noida, India

- Created a Resource Monitoring System (Cloud watch) Using Prometheus, Grafana and Alert manager to Send Alerts, Monitor Servers on Grafana Dash Boards. Link to Project- https://github.com/asking28/resource-monitoring/wiki
- Features developed include Sending Email alerts on Server Over-Usage, Monitoring through Prometheus and Grafana dashboards.

Honors, Awards, and Presentations

1. Secured First position in Financial Narrative Summarisation competition with highest ROUGE-1,L **2020**
2. Secured overall third position SemEval2020 Spanglish Sentimix task with F1-macro of 0.77. **2020**
3. Achieved highest score in post evaluation in Greek language offensiveness detection. **2020**
4. Samsung Spot Award for outstanding performance in the project. **2019**
5. Poster and Project Presentation at 100th Institute Day IIT (BHU). **2017**
6. Project Presentation at IIT Guwahati technical festival Techniche. **2016**
7. Awarded KVPY fellowship, Research Scholarship by Indian Government with top 99.4 percentile **2015**
8. Cleared IIT-JEE Advanced and IIT-Mains with top 99.99 Percentile. **2015**

Research Papers

1. Voice@SRIB at SemEval-2020 Task [9,12]: Sentiment and Offensiveness detection in Social Media - Accepted in SemEval(Coling2020).
2. PoinT-5: Pointer Network and T-5 based Financial Narrative Summarisation – Accepted in FNS2020.

   Under Review
3. Recommender System Based Research under review in ElseVier (Applied Soft Computing) Journal.

Projects

SemEval2020 task-9 Sentimix (Hinglish and Spanglish)

- Dataset contains social media comments and tweets on code-mixed languages such as Hindi+English, Spanish+English.
- Task was to classify the text into positive, negative, and netural sentiments. We achieved overall third ranking on Spanglish with F1-macro of 0.77.
- To clean and extract features from social media data, we applied multiple preprocessing steps such as Levenshtein distance based spell correction, contraction and acronym expansion, profanity vector, User and URL replacement.

- We used ensemble model of RNN based self-attention, CNN and LSTM based model with extensive hyper-parameter tuning.

SemEval2020 task-12 OffensEval (English, Greek, Turkish, Danish, and Arabic)

- English task was divided in three subtasks. A) Detect offensiveness, B)Targeted or not, C)Determine the target.
- For other languages mentioned, there was only one task- that is to detect Offensiveness in different language corpus.
- We used different preprocessing methods for different languages from CLTK library.
- We trained English corpus using pre-trained embeddings on Social-media corpus using Fasttext model. We trained using transformer based models on English language.
- We trained Turkish and Danish corpus using pre-trained embeddings using stacked ensemble model.
- We achieved F1-macro of 0.908 on English language and 0.886 on Greek language social media corpus.

Deep Learning based Academic Recommender System Asst.Prof.Sukomal Pal

- Recommendation of journals and conferences based on abstract and title of the research paper.
- We implemented CNN, LSTM and stacked ensembling on DBLP dataset and achieved highest top-15 accuracy of 75% using LSTM model, also created a web app for the same.

Object Recognition and Classification Asst.Prof.Bhaskar Biswas

- Object detection using YOLOv2 and YOLOv3 and YOLOv3_tiny architectures using Python and Tensorflow. Trained on Custom dataset and increased data using Tensorflow data Augmentation. [Project] https://github.com/asking28/Object-Recognition
- Evaluated Recall metrics of above architectures, with YOLOv3 being highest at 70% on testing images.

Electricity Price Forecasting Prof.S.P.Singh

- Price forecasting with Neural Networks on MATLAB, MAPE of 6.5%, and MAE of 5.2%.
- Analyzed hourly and weekly demands of Electricity Load and their corresponding prediction errors.

Relevant Coursework

Probability and Statistics	Mathematical Modeling	Data Mining
Calculus	Ubiquitous Computing	Mathematical Methods
Numerical Techniques	Machine Learning	

Extracurricular Activities

1. Won Gold Medal in Javelin Event in Spardha Sports Meet 2017, biggest sports festival in North India.
2. Served as Vice- Captain of Athletics Team at IIT(BHU) and lead the team at Inter-IIT Meet at IIT Madras.
3. Participated in Inter-IIT tournament at IIT-Kanpur and IIT- Madras for Javelin Event.
4. Participated in Blood Donation camps organized in college. Mentored Students with Under Privileged background.
5. Core team member of Drone Tech Event at Technex annual Techno-Management festival and Aero Modeling club of IIT (BHU).